

Yash Tiwari

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SKILLS

Programming Languages: Python

AI/ML: PyTorch, TensorFlow, scikit-learn, Transformers, LoRA Fine-tuning, LLMs (GPT-4o, Mistral, Qwen3-VL), RAG

Computer Vision: OpenCV, YOLO, DeepFace

Backend & Infra: FastAPI, Flask, PostgreSQL, pgvector, Redis, Qdrant, Docker, Ollama, vLLM

EXPERIENCE

Qoneqt

12/2025 – 02/2026

Software Engineer

Thane

- Developed AI agents for platform integrations and user verification, processing 20,000+ verifications daily.
- Built personalized AI agents for individual users, supporting 250,000+ users.

Qoneqt

06/2025 – 09/2025

Artificial Intelligence Engineer

Thane

- Architected and deployed a scalable Aadhaar authentication pipeline, raising verification accuracy from 48% to 95% and reducing manual review time by 80%.
- Deployed an LLM-powered mobile-app testing agent with Ollama and Maestro detecting 3× more critical bugs pre-release.

Google Summer of Code – SAT Montreal

05/2025 – 09/2025

Software Developer

Remote

- Architected high-performance real-time gesture processing system for professional musicians and live performers, porting Puara Gestures Library to Avendish runtime for low-latency live performance scenarios.

Stamp 'IT Robotai & Solutions Pvt. Ltd

03/2025 – 05/2025

Embedded Software Engineer

Thane

- Optimized deep learning inference pipeline on resource-constrained Raspberry Pi hardware achieving 30+ FPS YOLO object detection at 720p, while also developing a Qt-based desktop CAD application integrating real-time camera calibration, visualization tools, and automated geometric optimization workflows, reducing deployment setup time by 50%.

PROJECTS

Aadhaar AI Verification System | Python, FastAPI, Qwen3-VL, YOLO, DeepFace, Redis, Ollama

- Architected high-performance KYC system using Vision-Language Model (Qwen3-VL via Ollama) for document extraction and DeepFace for face matching, achieving 95%+ verification accuracy.
- Deployed as async FastAPI service with Redis-based caching and sub-second inference latency, reducing manual processing by 80% and handling 20,000+ daily verifications.

Qoneqt Agent Network | Python, FastAPI, GPT-4o, Qwen3-VL, vLLM, Qdrant, pgvector, Redis

- Built multi-component AI platform serving 250,000+ users: personalized chat agents (GPT-4o-mini with dynamic per-user prompts), RAG-based Memory Vault (Qdrant + sentence-transformers), and pgvector-powered recommendation engine with HNSW indexing.
- Designed LLM router with circuit breaker to route between external GPT-4o (user-facing) and local vLLM+Qwen3-VL (background post classification), optimizing cost and latency.

LLM-Powered Mobile App Testing Agent | Python, Mistral-7B, LoRA, Ollama, Maestro, Docker

- Fine-tuned Mistral-7B with LoRA on 10K+ instruction-YAML pairs to convert natural language test descriptions into Maestro test scripts for mobile apps.
- Integrated local LLM inference with Maestro workflows, detecting 3× more critical pre-release bugs.

EDUCATION

New Horizon Institute of Technology

Expected May 2026

B.E. in Mechatronics Engineering

Thane, Maharashtra

CERTIFICATIONS & AWARDS

Runner-Up, ClearRoute Agentic AI Hackathon (100+ teams) – Led 5-member team building a real-time racing analytics system with AI-powered strategy recommendations.

Robotics Club Lead – Led smart navigation system for visually impaired using sensor fusion; mentored juniors on embedded systems.

Certifications: IBM ML Professional (Coursera), Applied ML (U. of Michigan), CS Foundations (U. of London).